

Hiral Patel

Computer Engineering Lecturer | Machine Learning Educator

Machine Learning | Python | Data Science | Artificial Intelligence

Theory of Computation | Compiler Design

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Gujarat, India | Open to Relocation

[Teaching Portfolio](#) | [Research \(ORCID\)](#) | [GitHub](#) | [LinkedIn](#) | [YouTube](#)



Professional Profile

Passionate Computer Engineering educator and software developer with over **3 years of combined experience** in higher education and full-stack web development. I am currently pursuing an **M.Tech in Software Engineering** while serving as a Lecturer, delivering courses mainly in Python for Data Science, Machine Learning, Web Development, Theory of Computation, and Compiler Design. Experienced in teaching and creating digital learning resources for engineering students. Dedicated to bridging academic concepts with real-world industry practices through practical demonstrations and research-oriented teaching.

Education

Master of Technology (Software Engineering)

(2025–2027)

Gandhinagar University

Bachelor of Engineering (Computer Engineering)

(2014–2018)

Gujarat Technological University (GTU)

PTE Academic: 81/90

(Dec 2025)

Core Skills

Technical Skills

- Python
- SQL
- HTML5
- CSS3
- Bootstrap
- Machine Learning
- Data Visualization
- VSCode
- JavaScript
- PHP
- jQuery
- Laravel
- MySQL
- Streamlit
- GitHub

Academic Skills

- Student Mentoring
- Outcome-based Education

Career Summary

TEACHING EXPERIENCE (July 2025 – Present)

Organization : Neotech Technical Campus (GTU)

(July 2025 – Present)

Role : Lecturer

Subjects Taught

- Web Development (PHP)
- Database Management
- Theory of Computation
- Compiler Design
- Python for Data Science

Key Responsibilities

- Deliver lectures for diploma and undergraduate engineering students.
- Plan, prepare, and review lecture materials in advance, including examples, demonstrations, and practical activities, to ensure clear and effective delivery.

Faculty Development Programs (FDPs)

- **AICTE Training and Learning (ATAL)**
Academy Faculty Development Program on “Artificial Intelligence and Its Applications”
- **C-DAC** Faculty Development Program on “Artificial Intelligence and Computer Vision”

Teaching Philosophy

I believe engineering education should combine conceptual understanding with continuous practical implementation. My classrooms emphasize project-based learning, collaborative problem solving, and industry-relevant technologies.

Students' Feedback

[Teaching Feedback & Student Evaluations](#)

ACADEMIC PROJECTS

Student Performance Prediction

Python | Streamlit | Pandas | Plotly

Project:

https://github.com/hiral1547/Student_Analytics_Dashboard

- Built machine learning models to analyze academic performance.
- Performed data preprocessing, feature engineering, and visualization.
- Evaluated multiple algorithms for predictive accuracy.

Chatbot

Python | Artificial Intelligence | NLP

Project: <https://github.com/hiral1547/AI-chatbot>

- Developed an AI-powered chatbot using Python to provide interactive, automated responses to user queries.
- Implemented conversational logic and user interaction for an accessible question-answering experience.

Diabetes Prediction Dashboard

Python | ML | Streamlit | Data Visualization

Project: <https://github.com/hiral1547/Diabetes-Prediction-Dashboard>

- Developed a machine learning-based application for predicting the likelihood of diabetes using patient-related input parameters.
- Implemented data preprocessing and machine learning techniques for predictive analysis.
- Developed an interactive Streamlit dashboard for user input, prediction, and visualization.
- Designed the application to demonstrate the practical application of machine learning in healthcare analytics.

PUBLICATIONS & RESEARCH

ORCID: 0009-0007-8391-5466

DOI: <https://doi.org/10.32628/CSEIT2612416>

- Hiral Patel, *Machine Learning and Explainable Artificial Intelligence for Diabetes Prediction: A Systematic Review with Bibliometric Insights*, International Journal of Scientific Research in Computer Science, Engineering and Information Technology, 2026.

Professional Development

(Jan 2025 – Jun 2025)

Independent Learning & Skill Enhancement

- Prepared for college-level teaching sessions in technical subjects.
- Explored and practiced effective pedagogical techniques to improve student engagement and learning outcomes.
- Prepared for the PTE (Pearson Test of English) to improve communication skills and support potential international opportunities.

INDUSTRY EXPERIENCE (Mar 2022 – Dec 2024)

Role: Developer

Organization : SIIMTEQ Technologies

(May 2024 – Dec 2024)

Role : PHP/Laravel Developer

- Developed enterprise web applications using Laravel framework.
- Implemented authentication, authorization, and role-based access control.
- Collaborated with backend teams using Git version control.

Organization : SEPL – Suyog Electricals Ltd

(Aug 2023 – April 2024)

Role : PHP Web Developer

- Developed, fixed and changed functionality for web application.
- Imported Excel data to SQL Server Database.

Organization : Vtech Lab

(June 2023 – Aug 2023)

Role : Internship & Trainee

- Learned about PHP & Laravel.

Organization : Pavans Group

(March 2022 – May 2023)

Role : Frontend Developer

- Worked as frontend developer
- Developed responsive websites using HTML, CSS, JavaScript, Bootstrap, and jQuery.
- Improved website usability and mobile responsiveness.
- Collaborated with designers to deliver client-focused interfaces.

Digital Academic Presence

Teaching Materials Website

<https://lecture-materials.netlify.app/>

- Course materials and lecture notes
- Laboratory manuals

YouTube – RapidNotes

<https://www.youtube.com/@RapidNotes-23>

Educational channel providing:

- Computer Engineering Subjects' understanding
- Topic wise Notes on drive
https://drive.google.com/drive/folders/1Be3PZPJ2AK48_sufCQbITE7oaQ8m4iaW

Languages known

- English (Professional)
- Hindi (Fluent)
- Gujarati (Native)

References

Referee 1: [Dr. Pranavkumar D. Bhangaonkar |

Principal | NIT/NFDE |

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Referee 2: [Prof. Sanjay Macwan | Professor |

NIT/NFDE | sanjaymacwan.nfde@neotech.ac.in]

Referee 3: [Prof. Amit Patel | Professor | NIT |

amitpatel.nit@neotech.ac.in]